Weekly Progress Report

Project Title: Quantum Secure Email Client Application (QuMail)

Team ID: G14

Date: Week 2

Executive Summary

Current Sprint: Sprint 2 of 6

Sprint Goal:

The main goal of the second sprint was to perform detailed functional modeling of the Quantum Secure Email Client Application (QuMail). The focus was on finalizing the functional and non-functional requirements, identifying system actors and use cases, and defining the interaction between the User, QuMail, Key Manager, and Email Server.

Status: On Track

During this week, the functional requirements and major system interactions of QuMail were analyzed and documented. The primary use cases of the application were identified based on the requirements finalized during Week 1.

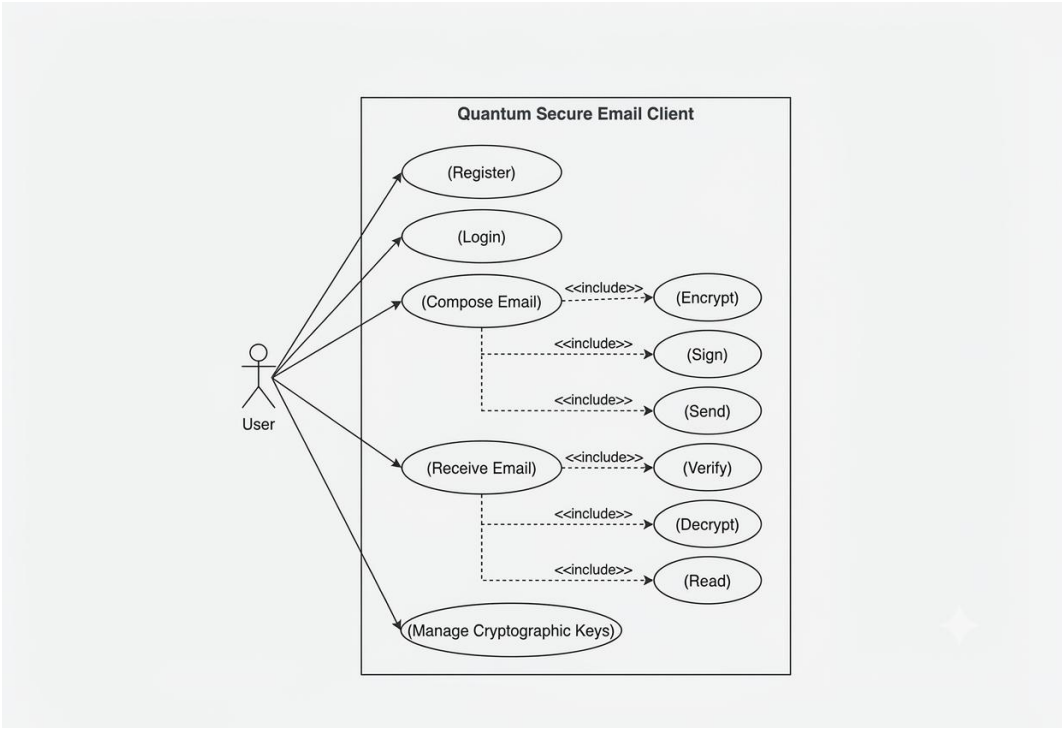
The major functions considered include user authentication, composing emails, secure encryption, key management, sending and receiving emails, email decryption, and integrity verification. These functions were organized to provide a clear understanding of the expected behavior of the system.

The initial UML modeling was also started during this week. A Use Case Diagram was prepared to represent the system boundary, actors, and major interactions with the QuMail application.

The overall system architecture was further discussed, focusing on the interaction between the User Interface, Email Processing module, Encryption/Security module, Key Manager interface, and Email Server.

UML & OOAD Progress

Diagrams Completed this week:



Use Case Diagram: The initial Use Case Diagram was prepared for QuMail. It represents the main actors and their interactions with the system.

- User Registration
- User Login
- Compose Email
- Encrypt Email
- Obtain/Generate Secure Key
- Send Email
- Receive Email
- Decrypt Email
- Verify Email Integrity
- View Inbox
- Logout

Diagrams Revised:

The overall system interaction and module responsibilities were reviewed based on the functional requirements.
The identified modules from Week 1 were further considered while defining the system interactions.

OO Concept Applied:

The object-oriented approach was applied by dividing the application into separate components with specific responsibilities.

- User Interface
- User/Authentication Module
- Email Processing Module
- Encryption/Security Module
- Key Manager Interface
- Email Service/Server Interface

Each module will handle its specific functionality, making the system easier to maintain, test, and upgrade.

Technical Execution (DevOps & Git)

Jira Tasks Completed:

- Functional requirements were reviewed and finalized.
- Non-functional requirements were identified.
- Major actors and system use cases were identified.
- Use Case Diagram was prepared.
- Major system interactions were analyzed.
- Key Manager and Email Server interaction requirements were studied.
- Major features were divided into smaller development tasks for upcoming sprints.
- GUI-related requirements for the email client were identified.

Git Activity:

- The existing Git repository was continued for project development.
- Project documentation and UML-related work were maintained.
- The repository structure was reviewed for upcoming development activities.
- Git will continue to be used for version control and collaborative development.

CI/CD Status:

CI/CD pipeline has not yet been implemented. It will be considered during the development phase after the initial application structure and codebase are created.

Testing Summary

Total Unit Tests: 0

Pass Rate: N/A

Key Bug

Identified:

No implementation-related bugs were identified during Week 2 because the project is still in the requirements and system modeling phase.

The main challenge during this week was defining how the Key Manager and encryption/security layer will interact with the existing email communication workflow. The system needs to maintain normal email functionality while providing an additional security layer for encryption, key management, and decryption.

Goals for Next Week

- Finalize the Use Case Diagram and other initial UML diagrams.
- Prepare the Class Diagram for the major QuMail components.
- Define the detailed system architecture.
- Finalize the interaction between QuMail, Key Manager, and Email Server.
- Start designing the GUI structure of the email client.
- Define the initial database/data requirements, if required.
- Study the Key Manager API integration requirements.
- Study the required email protocols and integration approach.
- Break down the major features into development tasks in Jira.
- Continue maintaining the project source code and documentation through Git.

Project Guide Signature _____ **Project In-charge Signature** _____